Contactless Vital-Sign Monitoring: Benchmarking Denoising Frameworks for mmWave Radar Signals

Malak Zriouli
Department of Data Science
Hanyang University
Seoul South Korea
zrioulimalak@hanyang.ac.kr

Yashu Shukla
Department of Data Science
Hanyang University
Seoul South Korea
yshukla@hanyang.ac.kr

Youngtae Noh
Department of Data Science
Hanyang University
Seoul South Korea
youngtaenoh@hanyang.ac.kr

ABSTRACT

Throughout the years, several studies have shown interest in leveraging the usage of millimeter wave(mmWave) radar, a contactless non-obstructive device used to capture fine-grained signals in objects and humans as well. Some of the paramount applications of the afore-mentioned radio frequency device is rooted in the healthcare sector, featuring the detection of vital signs namely heart rate and breathing rate. As we aim to tackle the noise interference problem that mmWave faces, we developed and evaluated three different methods to determine the most accurate and robust method in heart rate and breathing rate estimation. The first method is the simple baseline approach consisting of band-pass filtering (BPF) and Fast Fourier Transform (FFT) techniques. The second is Discrete Wavelet Transform (DWT) method and the third method relies on deep learning, composed of Convolutional Neural Network (CNN), Bidirectional Long Short Term Memory (BiLSTM) and attention layer in order to predict vital signs. The results highlighted the superior performance of the deep learning model, which achieved an MAE of 3.12 on heart rate and MAE of 2.99 on breathing rate predictions in comparison with both the first and the second method. This underscores the importance of mmWave in vital signs monitoring, where it can be used in clinical settings as a prevention tool of health complications in timely interventions.

CCS CONCEPTS

• Computing methodologies → Deep learning • Applied computing → Biomedical signal processing • Hardware → Millimeter wave radar

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
December, 2025, Seoul, South Korea
© 2025 Copyright held by the owner/author(s).

KEYWORDS

mmWave radar, Non-contact vital sign monitoring, Heart rate, Breathing rate, Deep learning model

ACM Reference format:

Malak Zriouli, Yashu Shukla and Youngtae Noh. 2025. Contactless Vital-Sign Monitoring: Benchmarking Denoising Frameworks for mmWave Radar Signals. In *Unpublished manuscript*, Department of Data Science, Hanyang University, Seoul, South Korea. *10 pages.* https://doi.org/10.1145/1234567890

1 Introduction

In order to measure vital signs including heart rate (HR) and breathing rate (BR), conventional methods require contact with the participants such as smart watches, chest belts and ECG electrodes [1,2]. In diverse monitoring cases, this has proven to be invasive, uncomfortable and inconvenient due to the contact nature of the devices. This is where millimeter wave (mmWave) radar comes into play (Figure 1). It provides a contactless non intrusive alternative where it can monitor and detect vital signs [3,4,5] from a distance at ease, making continuous monitoring more practical and convenient.

mmWave radar is a device using radio frequencies that can traverse objects such as clothes thanks to its millimeter wavelength property, ranging from 1 to 10 millimeter, hence the name millimeter wave. Once the signal is sent via the transmitters, it hits the chest of the person and goes back to the mmWave. The difference of the transmitted and received signal reflects the chest displacement, allowing the measurement of the vital signs through the phase difference extraction [3,4]. In terms of the mmWave applications, it has a wide range of applications. Besides monitoring vital signs, it was also applied in stress monitoring where they distilled human stress based on the subject's daily activities [6], and tracked stress during night time [7]. Furthermore, some researchers took interest in monitoring driver's fatigue [8] and driver's vital signs [9] to prevent any potential accidents. Another application included lie detection based on the breathing rate that varied based on how truthful the

person was [10]. Human localisation [11] mainly in smart homes as well as intruder localization locating [12] and gait identifying [13] was also one of the intriguing applications of mmWave. Nevertheless, the challenge that has long faced mmWave radar is noise interference [14,15] such as motion artifacts which consequently distorts the signal, making vital signs hard to extract.

In this paper, we aim to address the noise interference problem through the benchmarking of three different denoising methods with the goal of identifying the most noise-resilient and accurate method in the estimation of heart and breathing rate. The first method is a baseline classic method relying mainly on FFT analysis [16,17]. The second method is the Discrete Wavelet Transform (DWT) method combining diverse advanced signal extraction methods [18]. Finally, the last method is a deep learning model consisting of CNN, BiLSTM and an attention layer, where we include two denoising strategies and later on determine which yielded better results.



Figure 1: **AWR1642 EVM - DCA1000- AWR1642 and DCA1000 in the designed radar holder (from right to left)**

2 Literature Review

Diverse researchers have been interested in the potential of mmWave radar in monitoring vital signs, mainly HR and BR. Hence they developed and adapted diverse techniques, including both traditional and advanced approaches in order to contribute to the evolution and enhancement of monitoring systems. For instance, Brandstetter et al. [18] proposed a two-stage Frequency-Modulated Continuous Wave (FMCW) radar method for HR estimation for in vehicle application. The authors acknowledge that the dataset was collected from only a single participant, and thus limited the dataset size to only 74 measurement sessions which were collected in real-world driving across diverse traffic scenarios, summing to a total of 111 minutes of recording. The paper had divided their HR estimation approach in two main parts, the first stage involved using discrete wavelet transform (DWT) and autoregressive (AR) spectral analysis and the next stage covered relevance vector machine (RVM) approach. With this novel approach they achieved relative heart rate estimation error of approximately 6.9%. Another reliable HR estimation study using mmWave was carried out by Zhao et al. [20] on participants seated in various poses. The dataset was collected using TI IWR6843 mmWave FMCW radar from two participants across multiple sitting and lying poses. Data was collected for a total of eight poses, in which four were sitting poses and four were lying poses. The paper implemented a novel deep learning approach(DNN) and achieved an overall accuracy of 95.26%, with per pose accuracy ranging from 91.08% to 97.89%.Furthermore, Iyer et al. (2022) were particularly interested in detecting arrhythmia through the usage of machine learning [22]. Thus, they trained an artificial neural network on MIT-BIH database and tested on 15 subjects collected via TI IWR1443 mmWave FMCW radar. They could achieve a mean test accuracy of 75%, having MAE= 6.1bpm for HR and MAE=3.32 bpm for BR in standing position at 40cm, and MAE= 5.36 bpm for HR and MAE=2.56 bpm for BR in standing position at 40cm. They also mentioned that the HR values captured by the mmWave radar diverged from the reference by ±4. Thus, one should keep in mind that there is an error margin in the mmWave radar received signals as well. Moreover, Yen et al.(2022) were interested in deploying their monitoring system to be realistically used in medical settings having hospitalized older patients as a target. They followed two steps; first they removed unsatisfactory signals using a support-vector machine (SVM), the second step consisted of extracting both HR and BR through a singular value decomposition method. Their dataset comprised 10 healthy subjects data collected in laboratory settings through the usage of CW 24 GHz Doppler radar, which is a type of mmWave radars. The authors' results showed an accuracy ranging from 93.2% to 100%, where the HR and BR errors were between 0 to 4 bpm and 0 to 1 bpm respectively. Eventually their system was implemented in a clinical setting thanks to the strong performance of their models. Besides monitoring one participant at a time, Benny et al. (2024) study explored an innovative possibility to estimate the HR and BR of multiple subjects simultaneously through FMCW Radar IWR1642BOOST [23]. Their dataset had 400 samples with a maximum of 5 subjects and was not inclusive of diverse physiological conditions. Also, their regression models were trained on 80% of the full dataset. In terms of the models, they implemented diverse regressors inclusive of linear regressor, support vector regressor, K-nearest neighbors regressors and random forest (RF) regressor. The results were quite impressive indicating that RF regressor outperformed the competing models where their MAE for BR ranged between 0.39 to 1 and MAE for HR oscillated between 3.64 and 9.54, with RF regressor achieving the lower bound of the performance ranges. Also, they could monitor up to 5 subjects at the same time. According to the authors, their ultimate ambition is extending beyond 5 subjects to an arbitrary number of people, which would boost the efficiency of vital signs monitoring systems.

Based on the stated related works and relevant studies not explicitly mentioned here, there were not numerous academic papers and studies that investigated the optimal denoising approach by comparing multiple methods including both traditional and deep learning based approaches on a dataset with diverse participants having different physiological conditions and scenarios. To address these gaps, this study systematically evaluated three distinct signal denoising and vital sign extraction/prediction approaches: classical signal processing, Dwt-Kalman-AR method, and a deep learning model using the same mmWave radar dataset with the goal of determining the most robust and exact approach in estimating vital signs.

3 Methodology

3.1 Dataset

The dataset employed in this paper is a second-hand dataset titled 'mm-Wave FMCW Radar Vital Sign Monitoring Dataset: Diverse Physiological Scenarios' [19]. It consists of the heart rate and respiratory rate vital sign signals from mm-wave Frequency-Modulated Continuous Wave (FMCW) radar (AWR1642BOOST), and had as a goal the facilitation of research in non-invasive health monitoring.

The dataset encompasses 10 participants, with a variety of sex, age, weights and heights where men and women's ratio was perfectly balanced at 5 participants each, the average age and weight were 30.2 ± 6.32 years and 68.1 ± 14.14 kg, respectively (Table 1).

There were diverse scenarios that featured resting, elevated heart rate activity, asthma and meditation which depicted the extreme physiological cases. Also, the ground truth for heart rate was captured using the chest belt Polar H10 sensor. For breathing rate, the participants were instructed to count their breath meticulously, which was used as reference (Table 2).

In terms of the scenarios (Figure 2), there are 4 cases which are as follows: Distance (40cm, 80cm, 120cm, 160cm), Orientation (front, back, right, left), Angle (0 deg-80cm, 30deg-80cm, 45deg-80cm) and Elevated (80cm). Each case had 4 recordings of 62 seconds each, resulting in 44 binary files per resting participant and 48 per participant having participated in elevated heart exercise. Thus, the total of the used files are 456 binary raw file.

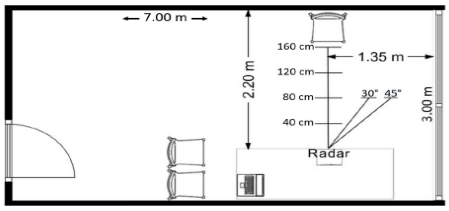


Figure 2**: Floor map dataset collection scenarios**

Attribute	Details	
Total Participants	10 (5 male, 5 female)	
Age / Weight (mean ± SD)	30.2 ± 6.32 years / 68.1 ± 14.14 kg	
Conditions	Normal Sitting (n=10) └ Asthma (n=2) └ Meditation (n=1) └ Elevated Exercise (n=4)	Participant 1~10 └ Participant 5,6 └ Participant 2 └ Participant 2,3,4,6

Table 1**: Participants details and conditions**

Scenario	Description	Files/Cases	Total
Distance	40 cm, 80 cm, 120 cm, 160 cm	4 files * 4 cases	**Total files** 456 files **Total Recording Time** 7h 51 mins 11s **1file = 1min*
Orientation	Front, Back, Left, Right	4 files * 4 cases	
Angle	0 deg, 30 deg, 45 deg	4 files * 3 cases	
Elevated	Elevated Exercise	4 files * 1 case	

Table 2**: Distribution of Scenarios**

Parameter	Value
Start Frequency	77 GHz
End Frequency	81 GHz
ADC Start time	6 μs
ADC Samples	250
Sample Rate	6250 ksps
Ramp End Time	50 μs
Idle Time	7 μs
Frame Count	1200
Frame Periodicity	50 ms
Chirp Loop Count	128
RX Gain	30 dB
TX Count	2
RX Count	4

Table 3: **Configuration parameters of AWR1642BOOST**

3.2 Signal Preprocessing

The signal preprocessing steps are shared across all methods, which includes important procedures to ensure noise suppression and improved signal quality. The preprocessing started by organizing the raw data to the required format, loaded the raw binary radar data and converted the interleaved I/Q (In-phase and quadrature components) data to complex numbers, and then reshaped it into frames, chirps, ADC samples, and virtual antennas, in accordance with the same "AWR1642BOOST" radar configuration used during data collection (Table 3). The next step focused on detecting the range bin, in our case the

interest of the region is the chest area, which contains the main information for HR and BR. The dataset includes multiple scenarios with different cases, to ensure reliable chest motion signal extraction, the range bin was carefully adjusted for each setting. This step was crucial because even small misalignments in the range bin can lead to capture of noise from surrounding background reflections, which directly affects the quality of the extracted physiological signal. As individual antennas capture multi path reflections and are subject to angle and position dependent noise, in order to obtain clean physiological signals from the radar we applied antenna fusion techniques which averages the phase signals across all virtual antennas for each frame. After that, the fused signal was detrended to eliminate low frequency drifts and unwrapped to fix discontinuities. Next, bandpass filtering was performed in accordance with standard human physiological rates, reflecting typical adult breathing and normal resting heart rates, focusing on respiration with the range of 0.1 - 0.6 Hz and heart rate with the range of 0.8 - 2.0 Hz. This filtering stage ensures that the motion artifacts and high frequency noise are suppressed, and only the relevant physiological motions are isolated.

3.3 Classical Baseline Approach

The classical baseline method was applied consistently across all four scenarios of distance, orientation, angle and elevated. On the preprocessed signal the HR and BR were estimated using two complementary techniques: spectral estimation and time domain peak analysis. Both approaches were utilized to increase robustness, as spectral analysis performs well under stable signals while peak-based estimation remains reliable under moderate noise. Spectral estimation via periodogram, applies the FFT and identifies the dominant frequency within each physiological band, whereas time-domain peak analysis counts respiratory and cardiac cycles relative to the recording duration. Finally the method outputs the extracted HR and BR for each recording. The classical baseline approach was implemented to establish a reference and have better understanding of the radar data. While this method provides reasonable estimates of HR and BR, it has some limitations too. The pipeline of steps does not completely address the problem of noise and subtle variations in the radar signals. The architecture of the classical method is illustrated in Figure 3 .

3.4 DWT-Kalman-AR Approach

The initial signal preprocessing was applied the same as mentioned in the section. In order to isolate the frequency components relevant to HR and BR, Discrete Wavelet Transform (DWT) was applied with a db4 wavelet and five levels. The Daubechies 4 wavelet (db4) was carefully

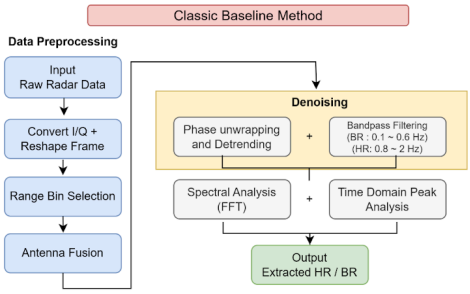


Figure 3: **Classical baseline approach architecture**

selected as it is well known for its effectiveness in signal processing, the number 4 represents the four vanishing moments, which provide an optimal balance between capturing smooth trends and preserving rapid variations, ensuring reliable isolation of heart rate and breathing rate components from chest displacement signals. Each of the five levels capture different frequency bands of the chest-displacement signal, allowing selection of those corresponding to heart rate and breathing rate and at the same time ensuring to filter out unrelated noise. For HR extraction, detail coefficients corresponding to the physiologically frequency range (0.8 - 2.2 Hz) are selected, and all other coefficients were zeroed. Whereas, the slower frequency components, which correspond to around 0.08 to 0.33 Hz, are used for breathing rate (BR) extraction. The breathing range in here was more narrow than that used in the classical baseline method, this is because DWT decomposition isolates slower oscillations more effectively, providing a cleaner estimate of respiration. To further enhance the signal quality and improve HR estimation, the reconstructed HR signal was smoothed using a sequential one-dimensional Kalman filter, which reduces random fluctuations. To the smoothed signal an autoregressive (AR) spectral model was applied, where the AR coefficients were computed via the Yule-Walker method, and the power spectral density (PSD) was analyzed to identify the peak frequency within the HR band. The peak frequency was then converted to beats per minute (bpm), hence printing the final HR estimate. Since breathing is slower and more regular than HR, an autoregressive model was not utilised for BR because reliable estimation can be made directly from peak intervals. So for breathing rate, after the DWT decomposition, a peak detection method was applied, in which the algorithm identified local maxima in the signal that correspond to individual breaths, while maintaining a minimum time interval between peaks to avoid false detections. The final BR estimate was then printed by converting the total number of observed peaks to breaths

per minute (bpm). Figure 4 shows the flowchart of the dwt-kalman-ar method.

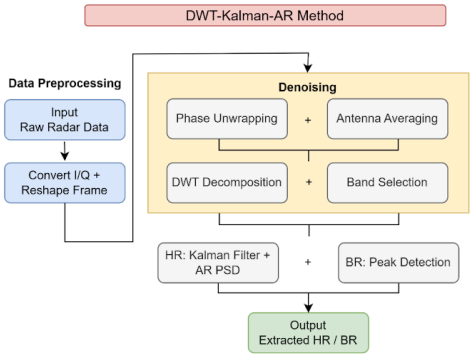


Figure 4**: DWT-Kalman-AR approach architecture**

3.5 Deep Learning Approach

The initial signal preprocessing was applied in the same manner as described in the previous section. The main modeling part starts from preparing the model input signal, in which signals were segmented into overlapping 10 second windows with 1 second steps, ensuring each window accurately included both amplitude and phase channel. To improve generalization and ensure an increased amount of data variety, data augmentation was applied by including random amplitude scaling, Gaussian noise addition, and temporal shifts. In terms of HR prediction, the model architecture consisted of CRNN+BiLSTM. First the signal passed through three 1 dimensional convolution layers having increasing filter sizes (32 → 64 → 128), which extracted local temporal features from amplitude and phase sequences. Each layer was followed by Batch Normalization and MaxPooling. Residual connections between convolution blocks ensured that the important features were carried forward. After that the processed signal was passed through two Bidirectional LSTM layers which captured long term dependencies and temporal dynamics in chest displacement signals. The HR output was then passed through layer normalization along with few dense layers with dropout to prevent overfitting. For BR prediction, the model architecture incorporated CNN+BiLSTM+Bahdanau attention. The CNN block started by identifying and extracting the temporal and local features of the breathing signal. It included Batch Normalization to improve generalization across the different participants and scenarios. It was followed by average pooling layers. Next, the extracted features were passed to the BiLSTM block, enabling the detection of the rhythmic structure of the breathing signal. Finally, the signal went through the Bahdanau attention layer where it focused on the most informative and relevant breathing cycles' signal portion while ignoring noise interference, further improving the model's performance. The attention mechanism also included a dropout layer in order to prevent overfitting. Finally, the BR was estimated through the fully connected layers. The model was trained using a Huber loss function and optimized with an Adam optimizer for Hr and AdamW optimizer for BR, both featuring a cosine decay learning rate schedule. All standard training techniques such as early stopping, learning rate reduction on plateau, and model checkpointing were included as well. Finally the output predicts HR for each input window, providing an accurate and robust estimation of heart rate in varied experimental scenarios. For the BR output, it was predicted for a full recording sequence. Testing : To assess the robustness of the proposed model, we evaluated it across multiple physiological and activity scenarios. For heart rate estimation, the model was tested under normal sitting, asthma participant conditions, and elevated exercise. For breathing rate estimation, the model was tested under normal sitting, meditation, and elevated exercise conditions. The test set was carefully designed to evaluate the model's ability to generalize under varying motion and physiological dynamics. The testing was carried out on unseen mmWave radar recordings using a sliding-window approach. In terms of HR predictions, the preprocessed amplitude and phase signals were segmented into overlapping 10 second windows with a 1 second step. However, for the BR predictions, there was only one ground truth value given per recording, thus the whole sequence was processed without windowing. Using the inverse of the training scaler, the predicted values were then transformed back to the original scale. To account for systematic prediction offset, a global bias was computed as the mean difference between predicted and ground-truth heart rate values across all test windows and subtracted from all predictions. Model performance was then evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE) , Root Mean Squared Error (RMSE). The architecture of the deep learning method is shown in the form of a flowchart in Figure 5.

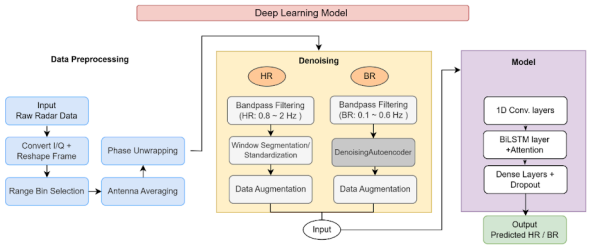


Figure 5: **deep learning approach architecture**

4 Results

The table 4 summarizes the result metric across all the three methods.

Given that MAE estimates directly how many beats per minute the model diverges from the ground truth, it is stable, intuitive and has a clinical meaning in vital signs predictions, which is the reason behind choosing MAE as the main evaluation metric.

The classical baseline method yielded the lowest accuracy, with an MAE of 9.95 bpm for HR and 12.9 bpm for BR, indicating limited robustness under varying physiological conditions. The second approach, the DWT-Kalman-AR method showed improved estimation accuracy, lowering HR MAE from 9.95 to 7.96 bpm and BR MAE from 12.9 to 6.25 bpm, along with notable reductions in MSE and RMSE. The deep learning model achieved the best performance in ever evaluation metrics. The DL model obtained a MAE of 3.12 bpm, with an MSE of 15.75 bpm and an RMSE of 3.97 bpm for HR estimation. Similarly, for BR estimation, it achieved an MAE of 2.99 bpm, an MSE of 11.4 bpm, and an RMSE of 3.38 bpm. The results clearly demonstrated (Figure 6) that the deep learning framework outperformed the former two methods with better predictive accuracy and robustness compared to other vital sign estimation methods.

	Vital Sign	MAE	MSE	RMSE
Baseline Method	HR	9.95	169.99	13.04
	BR	12.9	197.49	14.05
DWT-Kalman-AR Method	HR	7.96	98.69	9.93
	BR	6.24	69.89	8.36
Deep Learning Model	HR	3.12	15.75	3.97
	BR	2.99	11.44	3.38

Table 4: **Performance results of the three denoising approaches**

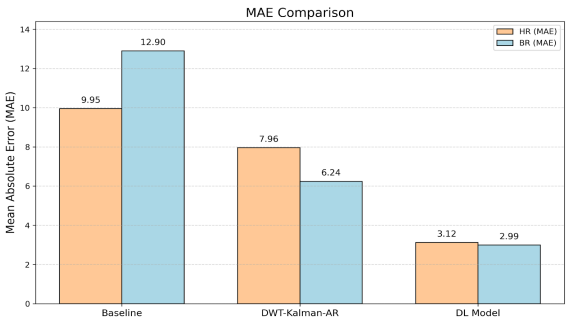


Figure 6: **Graph depicting performance results of the three denoising approaches based on MAE**

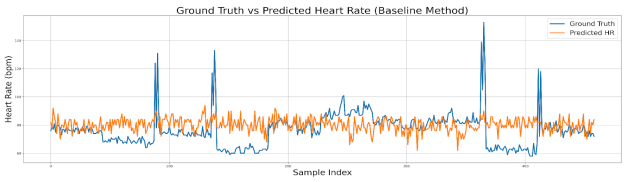
The following graphs are the descriptions of the plots that illustrate a comparison between the estimated and ground truth HR and BR vital signals for all the three methods. As seen in Figure 7, the predicted line graph shows low variability. For both HR and BR, the baseline method is able to follow the general ground truth trend during stable periods, but due to its strong smoothing effect, it prevents it from capturing the full physiological dynamic range. This results in consistent under estimation of peak heart rates and breathing rates, showing it could not accurately adapt to rapid physiological changes.

As seen in Figure 8, the DWT- Kalman- AR Method shows a significant performance improvement over the baseline approach. For HR, the method successfully captures the full dynamic range and accurately estimates the sharp peaks. In the BR plot, it shows improved accuracy for lower breathing rates and performs well during stable periods, however, it still struggles to follow the higher Ground Truth BR values.

The deep learning HR test prediction graphs are shown in Figure 9. In the normal sitting conditions(graph 1 and 3 of Figure 9), the predicted HR closely follows the Ground Truth pattern, capturing the moment-to-moment variations and successfully tracking most fluctuations across the stable range, hence demonstrating accurate prediction.

In the asthma elevated scenario (graph 2 of Figure 9), the predicted HR accurately captures the lower HR levels characteristic of an asthmatic patient, and in the elevated exercise phase (highlighted region), the model correctly adapts to the sharp increase in HR, showing strong responsiveness to sudden and effective tracking of high-intensity HR rates.

In terms of BR predictions (Figure 10), we can notice that the model could generally follow the trend of the main BR pattern through time, which is satisfactory. However, the predictions still diverged from the ground truth mainly in the first and second graph. This can be explained through the fact that there was only one ground truth per recording for the BR case, unlike the HR where the ground truth was given each second, expanding the limited dataset and allowing the model to predict windows of time instead of a full recording. Although data augmentation was performed, this was still a limitation especially in special case where the samples were restricted to 1 or 2 participants, not allowing the model to generalize to special physiological cases.



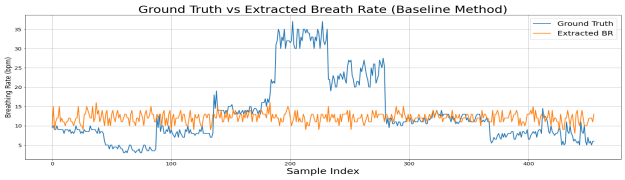


Figure 7: **Classical Baseline HR and BR estimation results**

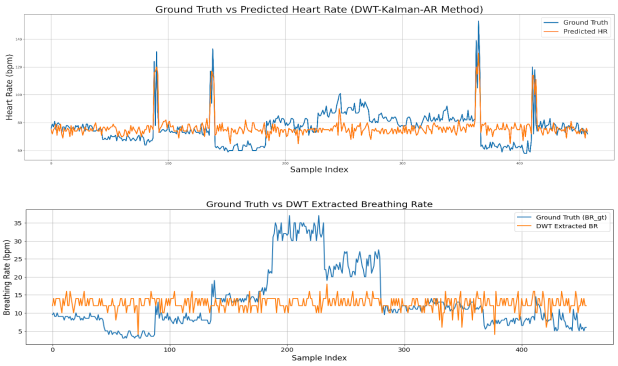


Figure 8: **Dwt-Kalman-Ar HR and BR estimation results**



Figure 9: **DL Model HR prediction results**

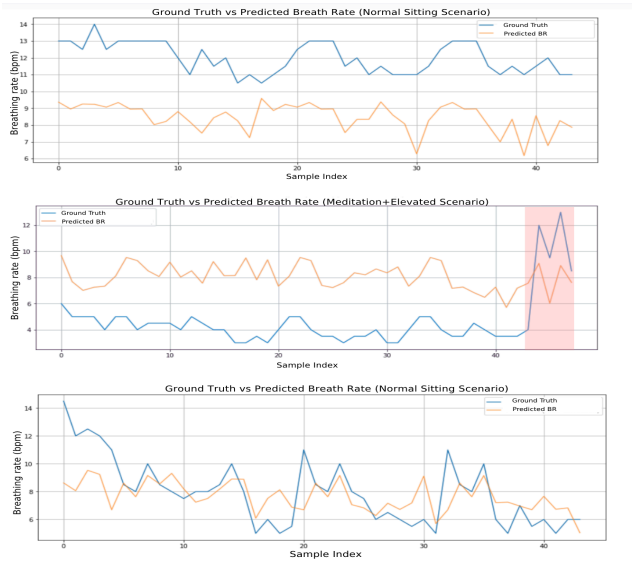


Figure 10: **DL model BR prediction results**

5 Limitations and Future Work

While we could extract the vital signs from the raw data of mmWave radar with satisfactory performance, there have been several challenges that are yet to be tackled. First, the dataset included diverse conditions such as normal sitting, asthma, meditation and elevated exercise. Nevertheless, we have a problem of class imbalance. Although we performed data augmentation and included special cases in the model validation process to attempt to address this problem, the model could not generalize well to some cases due to the learnt bias from the majority data, and the lack of enough instances from the same case. For instance, in terms of the meditation case and asthma, we only had one and two participants respectively. Thus, there were not enough samples for the model to be robustly generalized to those special cases. Nonetheless,we can say that the noise was effectively captured regardless, and the model could still learn the main patterns of the vital signs given that it could still follow the trend over time. Furthermore, pretraining the model on larger dataset could have improved its performance, but due to the lack of a standardized vital signs datasets captured by mmWave radar, it proved to be limiting considering that if we were to train the model with another dataset not following the same software configurations as the used mmWave radar, the model could wind up discarding some important signals due regarding them as noise, which explains the reason behind discarding the pretraining possibility. In addition, given that the dataset is relatively small, overfitting was bound to

happen. Therefore, to tackle it, we applied diverse prevention practices inclusive of model complexity control, regularization such as adding dropout layers, early stopping,noise injection through data augmentation and scheduling learning rate.

To enhance our work, our future plan includes expanding the dataset in terms of instances and conditions. This would have the objective of enabling the model to generalize better to diverse physiological cases. Also, we are also considering developing other deep learning models with denoising properties that work well with time series data then stacking them to check how the performance can improve. Finally, we are also looking forward to exploring monitoring multiple subjects at once to contribute to the clinical contactless assessment's efficiency. Our ultimate goal is to deploy a deep learning model in clinical settings to continuously monitor and detect abnormal vital signs as a step of a precaution monitoring system allowing the early detection of diseases and their prevention thereof.

6 Conclusion

Through this project, we contributed to the mitigation of the noise interference problem facing mmWave radar signals, where we benchmarked three denoising frameworks including classical signal processing relying on FFT, DWT-Kalman-AR method as well as deep learning based approach, based on a dataset including diverse physiological conditions and various scenarios. The classical signal method incorporated FFT spectral analysis and peak detection in the bands of vital signs. However, this method was sensitive to noise and could not fit the elevated heart and breathing rate cases. In terms of the second method, it relied on wavelet denoising combined with both kalman filtering and AR spectral estimation, which allowed for a better performance than the first method because it handles non-stationary signals and suppresses clutter and noise better. Nevertheless, this method depends on the rules of manual design of signal decomposition, which does not generalize well across subjects and is sensitive to some configurations of the mmWave radar. This brings us to the third method, which is the deep learning approach incorporating CNN, BiLSTM and attention layer that included two denoising strategies, one relying solely on bandpass filtering and the second included denoising autoencoders. We conducted an ablation study to check the effect the autoencoder denoiser would have on the vital sign waveform, hence on the overall performance of the model. The performance results showed that the model delivered better results without the denoising autoencoder (DAE). This suggests that the model must have filtered out the noise already thus by adding DAE, it was redundant and became counterproductive. Nevertheless, the model with the DAE included still did better than the former two methods where our results proved that the deep learning model - CNN+BiLSTM+Attention - outperformed the above mentioned two methods, where it achieved HR MAE = 3.19 bpm and BR MAE = 2.99 bpm, which meets the standards of reliability of <4 bpm for radar-based monitoring, given that a numerous studies' results showed results below the aforementioned number, which was considered to be satisfactory. These results suggest that advanced neural models do have the potential to support the mmWave contactless alternative by delivering noise resilient and robust solutions in terms of HR and BR predictions.

According to the World Health Organization, cardio-vascular diseases (CVD) have been a principal root of death globally, where mortality reached 19.8 million people in 2022 due to CVD. Also, Pleasants et al. (2022) research showed that 39.1% of the adult population in the US suffered from respiratory complications such as asthma and chronic obstructive pulmonary disease [21]. This underlines the importance of contributing to the development of optimal vital signs monitoring systems, which falls under the scope of our project. Subsequently, our model can be deployed as a way to continuously monitor and support the early detection of such diseases in clinical settings. By doing so, we will have contributed to the prevention of potential health complications which can be done through timely interventions.

ACKNOWLEDGMENTS

We would like to thank our supervisor professor Noh Youngtae and researcher Shin Seungyong for their valuable guidance throughout this project. We would also like to extend our gratitude to the Department of Data Science for supporting this work.

REFERENCES

[1] Keene, C. M., Kong, V. Y., Clarke, D. L., & Brysiewicz, P. (2017). The effect of the quality of vital sign recording on clinical decision making in a regional acute care trauma ward. Chinese Journal of Traumatology, 20(05), 283-287.

[2] Zheng, Y. L., Ding, X. R., Poon, C. C. Y., Lo, B. P. L., Zhang, H., Zhou, X.L., Yang, G.Z., Zhao, N. and Zhang, Y.T. (2014). Unobtrusive sensing and wearable devices for health informatics. IEEE transactions on biomedical engineering, 61(5), 1538-1554. DOI: 10.1109/TBME.2014.2309951

[3] Yang, Z., Pathak, P. H., Zeng, Y., Liran, X., & Mohapatra, P. (2016, July). Monitoring vital signs using millimeter wave. In Proceedings of the 17th ACM international symposium on mobile ad hoc networking and computing (pp. 211-220). DOI:

[4] Xiang, M., Ren, W., Li, W., Xue, Z., & Jiang, X. (2022). High-precision vital signs monitoring method using a FMCW millimeter-wave sensor. Sensors, 22(19), 7543.

[5] Wu, Y., Ni, H., Mao, C., Han, J., & Xu, W. (2023). Non-intrusive human vital sign detection using mmWave sensing technologies: A review. ACM Transactions on Sensor Networks, 20(1), 1-36.

[6] Liang, K., Zhou, A., Zhang, Z., Zhou, H., Ma, H., & Wu, C. (2023). mmStress: Distilling human stress from daily activities via contact-less millimeter-wave sensing. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 7(3), 1-36.

[7] Xu, X., Zhang, D., Lu, Z., Chen, J., Wu, Z., Geng, R., ... & Chen, Y. (2025, April). Contactless nighttime stress monitoring with mmwave

radar. In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
[8] Juncen, Z., Cao, J., Yang, Y., Ren, W., & Han, H. (2023). mmdrive: Fine-grained fatigue driving detection using mmwave radar. ACM Transactions on Internet of Things, 4(4), 1-30.
[9] Wang, F., Zeng, X., Wu, C., Wang, B., & Liu, K. R. (2021). Driver vital signs monitoring using millimeter wave radio. IEEE Internet of Things Journal, 9(13), 11283-11298.
[10] Wang, Y., Gu, T., Luan, T. H., & Yu, Y. (2022, September). Your breath doesn't lie: multi-user authentication by sensing respiration using mmwave radar. In 2022 19th Annual IEEE International Conference on Sensing, Communication, and Networking (SECON) (pp. 64-72). IEEE.
[11] mmTrack: Passive multi-person localization using commodity millimeter wave radio
[12] Zhao, P., Lu, C. X., Wang, J., Chen, C., Wang, W., Trigoni, N., & Markham, A. (2019, May). mid: Tracking and identifying people with millimeter wave radar. In 2019 15th international conference on distributed computing in sensor systems (DCOSS) (pp. 33-40). IEEE.
[13] Yang, X., Liu, J., Chen, Y., Guo, X., & Xie, Y. (2020, July). MU-ID: Multi-user identification through gaits using millimeter wave radios. In IEEE INFOCOM 2020-IEEE conference on computer communications (pp. 2589-2598). IEEE.
[14] Xiang, M., Ren, W., Li, W., Xue, Z., & Jiang, X. (2022). High-precision vital signs monitoring method using a FMCW millimeter-wave sensor. Sensors, 22(19), 7543.
[15] A high precision vital signs detection method based on millimeter wave radar
[16] Wang, Y., Wang, W., Zhou, M., Ren, A., & Tian, Z. (2020). Remote monitoring of human vital signs based on 77-GHz mm-wave FMCW radar. Sensors, 20(10), 2999.
[17] Alizadeh, M., Shaker, G., De Almeida, J. C. M., Morita, P. P., & Safavi-Naeini, S. (2019). Remote monitoring of human vital signs using mm-wave FMCW radar. IEEE Access, 7, 54958-54968.
[18] Brandstetter, J., Knoch, E., & Gauterin, F. (2025). Heart rate estimation using FMCW Radar: A Two-Stage method evaluated for In-Vehicle Applications. Biomimetics, 10(9), 630. https://doi.org/10.3390/biomimetics10090630
[19] Sadeghi, E., Skurule, K., Chiumento, A., & Havinga, P. (2024). Comprehensive mm-wave FMCW radar dataset for vital sign monitoring: Embracing extreme physiological scenarios. arXiv preprint arXiv:2405.12659.
[20] Zhao, P., Lu, C. X., Wang, B., Chen, C., Xie, L., Wang, M., Trigoni, N., & Markham, A. (2020, May 1). Heart Rate Sensing with a Robot Mounted mmWave Radar. IEEE Xplore. https://doi.org/10.1109/ICRA40945.2020.9197437
[21] Pleasants, R. A., Heidari, K., Ohar, J., Donohue, J. F., Lugogo, N. L., Kanotra, S. M., Kraft, M., Mannino, D. M., & Strange, C. B. (2022). Respiratory Symptoms among US Adults: a Cross-Sectional Health Survey Study. Pulmonary therapy, 8(3), 255–268. https://doi.org/10.1007/s41030-022-00194-9
[22] Iyer, S., Zhao, L., Mohan, M. P., Jimeno, J., Siyal, M. Y., Alphones, A., & Karim, M. F. (2022). mm-Wave radar-based vital signs monitoring and arrhythmia detection using machine learning. Sensors, 22(9), 3106.
[23] Benny, J., Moudhgalya, N. N., Khan, M., Meena, H. K., Wajid, M., & Srivastava, A. (2024). Scalable multi-subject vital sign monitoring with mmwave fmcw radar and fpga prototyping. IEEE Sensors Journal.